

IVOLT

Privacy-First Technology & Security

Gmail IMAP Archiver

Operator & Configuration User Guide

Document version 1.0

Issued June 30, 2026

Classification: Confidential — Internal / Client

IVolt.Core.Email · IVolt.Core.Email.Gmail

Contents

1. System Overview
 2. Architecture & Data Flow
 3. Prerequisites
 4. Gmail Account Preparation (App Passwords)
 5. Configuration Files
 6. First Run — Load or Create a Configuration
 7. The Archiving Pipeline (End to End)
 8. Storage Layout & the Archive Pointer
 9. Indexing Model
 10. Attachment Handling & Zipping
 11. Throttling, Retries & Resumability
 12. Security Model
 13. Operating Procedures
 14. Troubleshooting
 15. Glossary
- Appendix A — Configuration Field Reference

1. System Overview

The IVolt Gmail IMAP Archiver is a console-driven utility that connects to a Gmail account over IMAP, retrieves every message together with its full metadata and attachments, and writes a durable, indexed, offline archive. It is designed for long-running bulk archival: connect once, then stream messages continuously under controlled throttling until the mailbox is fully captured.

The system preserves the exact wire octets of every message so that a cryptographic hash of the original can be recomputed at any time. This is an integrity guarantee: the archive certifies **what was received**, byte for byte. It does not assert anything about the truthfulness of message content — only its faithful capture and storage.

Design principle

The archiver certifies **provenance and integrity of capture**, never the truth of the content. Every stored message carries a SHA-256 of its raw bytes so tampering is detectable after the fact.

What the archiver captures

- **RFC 5322 message content** — headers, addresses, subject, dates, body (text and HTML).
- **MIME structure** — every part, decoded from base64 / quoted-printable, with attachments separated.
- **IMAP-layer metadata** — UID, flags, INTERNALDATE, RFC822.SIZE, MODSEQ.
- **Gmail proprietary extensions** — X-GM-MSGID, X-GM-THRID (thread id), X-GM-LABELS.
- **Integrity data** — SHA-256 of the raw octets, and the raw bytes themselves.

2. Architecture & Data Flow

The archiver is organized into cleanly separated responsibilities. Storage types are pure data containers; processing is stateless static logic that operates on those containers. This separation keeps the message model serializable and library-agnostic.

Component	Namespace / Type	Responsibility
Configuration container	IVolt.Core.Email.Configuration.Configuration_Definition_Container	Typed storage of account, server, performance, archive and index settings; JSON round-trip.
Configuration manager	IVolt.Core.Email.Gmail.Gmail_IMAP_Management_Classes	Prompt for email, locate config, DPAPI unprotect, load; interactively create when absent.
Message container	IVolt.Core.Email.Gmail.Gmail_IMAP_Message_Container	Pure storage for one message: raw bytes, headers, addresses, MIME parts, IMAP + Gmail metadata.
Message processor	IVolt.Core.Email.Gmail.Gmail_IMAP_Message_Processors	Static parse pipeline: split headers/body, decode MIME, assign body roles, hash raw octets.

End-to-end flow

```

Email prompt
  |
  v
Locate config -> Resources\Configurations\{sanitized}.json
                (DPAPI-protected, CurrentUser)
  |
  v
Unprotect + deserialize -> Configuration_Definition_Container
  |
  v
Connect IMAP (imap.gmail.com:993, SSL)
  |
  v
For each message UID:
  FETCH (raw octets + FLAGS + INTERNALDATE + X-GM-*)
    -> Process(byte[])      parse + SHA-256
    -> ApplyImapMetadata()  attach UID/flags/labels
    -> Store (JSON record, BYMONTH)
    -> Index (fields per config)
    -> Zip attachments (per-message ZIP)
    -> throttle / pause

```

3. Prerequisites

Requirement	Detail
Operating system	Windows. Configuration protection uses DPAPI (ProtectedData), which is Windows-only.
.NET runtime	.NET supporting System.Security.Cryptography.ProtectedData and Newtonsoft.Json.
Newtonsoft.Json	Serialization library used throughout (System.Text.Json is not used).
Gmail account	IMAP enabled, 2-Step Verification on, and a 16-character App Password issued.
Disk	Sufficient free space for raw archives plus zipped attachments (size the mailbox first).
Windows account	The archive must be run under the SAME user that created the config (DPAPI CurrentUser).

4. Gmail Account Preparation (App Passwords)

Gmail no longer permits basic password login for IMAP when 2-Step Verification is enabled. The archiver authenticates with a Google **App Password** — a 16-character credential scoped to a single application. This same value is used for the IMAP, POP and SMTP secret fields in the configuration.

- 1 Enable 2-Step Verification on the Google account.
- 2 In Google Account → Security → App passwords, generate a new app password.
- 3 Copy the 16-character value (Google displays it grouped in fours; spaces are ignored).

- 4 Provide it to the archiver when prompted — it is stored only inside the DPAPI-protected config.

Security note

The App Password grants mailbox access. Treat it like a key. It is never written to disk in plaintext by this system — it lives only inside the DPAPI-protected configuration file. Revoke it in the Google console if a workstation is lost.

5. Configuration Files

Each Gmail account has one configuration file. Files live under:

```
Resources\Configurations\{email_sanitized}.json
```

The **sanitized** name is the address lower-cased with every @ and . replaced by an underscore. For example:

```
markalicz@gmail.com -> markalicz_gmail_com.json
```

The file is stored **DPAPI-protected** (Windows Data Protection API, CurrentUser scope) and Base64-encoded on disk. It is unreadable outside the account that created it and cannot be copied to another machine or user and decrypted. On load, the manager reverses this: read Base64, DPAPI-unprotect, then deserialize into the typed container.

Template (unprotected authoring form)

The distributed template uses **###TOKEN###** placeholders. Tokens are resolved to concrete values when a configuration is created, and the resolved result is what gets protected and saved.

Token	Meaning
###EMAIL###	Full account address (used for login and SMTP username).
###PASSWORD###	Account / app password.
###APPLICATION_SPECIFIC_PASSWORD###	Google 16-character App Password.
###BASEPATH###	Root folder under which archives and indexes are written.
###EMAIL_SANITIZED###	Sanitized address used as a per-account subfolder name.

6. First Run — Load or Create a Configuration

On launch the manager prompts for an email address, sanitizes it, and looks for the matching configuration file.

Case A — configuration exists

- 1 The file is read and DPAPI-unprotected under the current Windows user.
- 2 The JSON is deserialized into `Configuration_Definition_Container.FromJson`.
- 3 The loaded configuration is returned to the archiving pipeline.

If decryption fails

A `CryptographicException` on load almost always means the file was protected under a different Windows account or on a different machine. Re-create the configuration under the account that will run the archiver.

Case B — no configuration

The manager runs an interactive wizard at the command prompt. It requests the App Password (masked input), the base archive path, and operational parameters, each with a sensible default in brackets. Gmail server values are pre-filled. The App Password is fanned out to the IMAP, POP and SMTP secret slots. The completed container is serialized, DPAPI-protected, and written to the sanitized path.

```
Email address: markalicz@gmail.com
No configuration found for 'markalicz@gmail.com'.

=== New Gmail configuration ===
Press Enter to accept the [default] shown in brackets.

Gmail app-specific password (16 chars): *****
Base archive path [C:\IVolt\Mail]:
Archive pointer starting increment [1212]:
Max emails per minute [100]:
...
Saved (DPAPI-protected, CurrentUser):
  Resources\Configurations\markalicz_gmail_com.json
```

7. The Archiving Pipeline (End to End)

Once a configuration is loaded, archival proceeds message by message. Each message passes through the same fixed sequence of stages.

#	Stage	Action
1	Connect	Open an SSL IMAP session to <code>imap.gmail.com:993</code> and authenticate with the App Password.
2	Enumerate	Select the mailbox (e.g. [Gmail]/All Mail) and list message UIDs to process.
3	Fetch	For each UID, <code>FETCH</code> the raw RFC 822 octets plus <code>FLAGS</code> , <code>INTERNALDATE</code> , <code>RFC822.SIZE</code> , <code>X-GM-MSGID</code> , <code>X-GM-THRID</code> , <code>X-GM-LABELS</code> .
4	Parse	<code>Process(byte[])</code> splits header/body, decodes MIME parts, assigns text/HTML body roles, computes SHA-256 of raw octets.
5	Enrich	<code>ApplyImapMetadata()</code> attaches UID, flags, labels and Gmail ids to the container.
6	Store	Write a JSON archive record into the <code>BYMONTH</code> folder structure; assign the next archive pointer.
7	Index	Emit an index entry for the enabled fields (body, subject, from, to, cc, bcc, date, attachments).

#	Stage	Action
8	Attachments	Extract decoded attachment parts and write them into a per-message ZIP.
9	Throttle	Honor max emails/minute and attachment MB/minute; pause the configured seconds; retry on transient failure.

Gmail label model

In Gmail, labels are exposed as IMAP folders and one message can appear under several. Archiving [Gmail]/All Mail captures each underlying message exactly once; the labels it carried are preserved in X-GM-LABELS rather than by duplicating the message per folder.

8. Storage Layout & the Archive Pointer

Archives, indexes and attachments are written under the configured base path, each in a per-account subfolder named by the sanitized email. The default structure is **BYMONTH**: messages are grouped into folders by the year and month of their date.

```
{BASEPATH}\
  ARCHIVES\{email_sanitized}\
    2026\01\ 2026\02\ ...      (BYMONTH message records, JSON)
  ATTACHMENTS\                (per-message attachment ZIPs)
  INDEX\{email_sanitized}\INDEX\ (index files, IVOLT_SCAN format)
```

The archive pointer

Each stored message is assigned a monotonically increasing **archive pointer** — a stable integer identity independent of Gmail's UID or message id. The first pointer is taken from **archive_pointer_starting_increment** (default 1212) and increases by one per archived message. The pointer links a stored message record to its index entry and its attachment ZIP, and gives every archived item a durable local key.

Why a separate pointer

Gmail UIDs are only unique within one mailbox and can shift with UIDVALIDITY changes; the archive pointer is controlled entirely by IVolt and never changes once assigned, so it is the safe cross-reference key for storage, index and attachments.

9. Indexing Model

Indexing produces a searchable record per message in the **IVOLT_SCAN** index format. Which fields are indexed is controlled by the `index_email_*` switches in the configuration, all enabled by default. Disabling a field keeps it in the stored archive record but omits it from the index — useful for reducing index size or excluding sensitive recipients.

Config switch	Indexes
index_email_body	Full text of the plain-text (and where present HTML) body.
index_email_subject	Subject line (RFC 2047 decoded).
index_email_from	From address and display name.
index_email_to / _cc / _bcc	Recipient address lists.
index_email_date	Message date for range queries and BYMONTH grouping.
index_email_attachments	Attachment filenames and content types.

10. Attachment Handling & Zipping

When `archive_attachments` is enabled, each attachment MIME part is decoded from its transfer encoding (base64 or quoted-printable) back to its original bytes, then written into a per-message archive in the configured attachments format (default ZIP). The ZIP is keyed by the message's archive pointer so it can be located directly from the stored record or the index.

- Filenames are recovered from Content-Disposition, falling back to the Content-Type name parameter, and RFC 2047 decoded.
- Inline parts referenced by Content-ID (embedded images) are preserved alongside true attachments.
- Attachment throughput is bounded by `max_attachment_size_in_MB_per_minute` to avoid saturating the link or tripping Gmail limits.

Integrity

Decoded attachment bytes are exactly those Gmail delivered. Because the parent message's raw octets are also stored and hashed, any attachment can be re-derived and checked against the original message if a dispute arises.

11. Throttling, Retries & Resumability

Bulk archival is a long process and Gmail enforces rate limits. The performance block governs pacing so a full-mailbox run completes without being throttled or disconnected by the server.

Setting	Default	Effect
max_emails_per_minute	100	Ceiling on messages fetched per minute.
max_attachment_size_in_MB_per_minute	50	Ceiling on attachment volume downloaded per minute.
pause_between_requests_in_seconds	11	Fixed delay inserted between requests.
max_retries	3	Attempts per message on transient IMAP/network failure before skipping.

Setting	Default	Effect
log_level	DETAIL	Verbosity of the operation log.

Because every stored message carries its archive pointer and Gmail message id, a run that is interrupted can resume: already-archived message ids are skipped, and the archive pointer continues from the last assigned value. Plan the first full capture during a window where the workstation can run uninterrupted; incremental runs afterward are short.

12. Security Model

- **Credential at rest:** the App Password lives only inside the DPAPI-protected config, bound to the Windows user and machine.
- **Credential in transit:** IMAP/POP/SMTP all use SSL/TLS to Gmail.
- **Credential in memory:** masked at the prompt so it never appears in console scrollback.
- **Archive integrity:** SHA-256 over raw message octets makes post-hoc tampering detectable.
- **Least exposure:** index fields for sensitive recipients (bcc) can be disabled without losing the stored record.

Portability boundary

DPAPI CurrentUser configs are intentionally non-portable. To move archiving to another machine or service account, recreate the configuration there. Do not attempt to copy the protected file across accounts — it will not decrypt.

13. Operating Procedures

Initial full archive

- 1 Prepare the Gmail App Password (Section 4).
- 2 Launch the archiver; enter the email; complete the new-config wizard (Section 6, Case B).
- 3 Confirm the base path has adequate free space for the whole mailbox plus attachments.
- 4 Start the run and let it proceed under throttling until enumeration completes.
- 5 Verify the ARCHIVES and INDEX trees populated, and spot-check a message record and its attachment ZIP.

Recurring incremental archive

- 1 Launch and enter the same email; the existing config loads automatically.
- 2 The run skips already-archived message ids and captures only new mail.
- 3 Confirm the archive pointer advanced and new BYMONTH folders appear as needed.

14. Troubleshooting

Symptom	Likely cause	Resolution
CryptographicException on load	Config protected by another user/machine	Recreate the config under the account that runs the archiver.
IMAP login rejected	Basic password used, or App Password revoked	Issue a fresh Google App Password; re-run the wizard to update the config.
Connection dropped mid-run	Gmail rate limiting	Lower max_emails_per_minute; raise pause_between_requests_in_seconds; resume.
Garbled subject or filename	Encoded-word / charset edge case	Confirm RFC 2047 decoding path; verify the source charset is installed.
Attachments missing from ZIP	archive_attachments disabled, or MB/min throttle	Enable archive_attachments; raise the attachment MB/minute ceiling.
Duplicate-looking messages	Same message under multiple labels	Archive [Gmail]/All Mail; rely on X-GM-MSGID for de-duplication.

15. Glossary

Term	Definition
App Password	A 16-character Google credential scoped to one application; required for IMAP under 2-Step Verification.
Archive pointer	IVolt-assigned monotonically increasing integer identity for a stored message.
BYMONTH	Storage structure grouping message records into year/month folders.
DPAPI	Windows Data Protection API; encrypts the config bound to a user or machine.
IVOLT_SCAN	The index record format produced during archival.
MODSEQ	IMAP CONDSTORE modification sequence used to detect changes.
Provenance	Verifiable record of how data was captured and stored — not a claim about its truth.
Sanitized email	The address lower-cased with '@' and '.' replaced by '_' for use as a filename.
X-GM-MSGID / THRID	Gmail proprietary 64-bit message and thread identifiers.
X-GM-LABELS	Gmail proprietary list of labels applied to a message.

Appendix A — Configuration Field Reference

Root fields

Field	Type	Purpose
email	string	Account address; IMAP login and SMTP username.
password	string	Account / app password (secret).
server	object	IMAP/POP/SMTP endpoints and secrets (below).
performance	object	Throttling and retry settings (below).
archive_index_folder	string	Destination for index files.
archive_folder	string	Destination for message records.
archive_structure	string	Grouping scheme, e.g. BYMONTH.
archive_format	string	Stored record format, e.g. JSON.
archive_attachments	bool	Whether to extract and store attachments.
archive_pointer_starting_increment	long	First archive pointer value (default 1212).
archive_attachments_format	string	Attachment container format, e.g. ZIP.
archive_attachments_folder	string	Destination for attachment archives.
index_fileformat_type	string	Index format, e.g. IVOLT_SCAN.
index_email_body ... _attachments	bool	Per-field index toggles (all default true).

server

Field	Default	Purpose
imap / port / ssl	imap.gmail.com / 993 / true	IMAP endpoint.
pop / pop_port / pop_ssl	pop.gmail.com / 995 / true	POP endpoint (optional).
smtp / smtp_port / smtp_ssl	smtp.gmail.com / 587 / true	SMTP endpoint.
smtp_authentication	true	Whether SMTP requires auth.
smtp_username	= email	SMTP login.
application_specific_password	(secret)	App Password for IMAP/POP.
smtp_password / smtp_application_specific_password	(secret)	App Password for SMTP.

performance

Field	Default	Purpose
max_emails_per_minute	100	Fetch rate ceiling.
max_attachment_size_in_MB_per_minute	50	Attachment download ceiling.
pause_between_requests_in_seconds	11	Inter-request delay.
max_retries	3	Retries per message on transient failure.
log_level	DETAIL	Log verbosity.

IVolt LLC — Privacy-First Technology & Security. This document is confidential and intended for authorized operators. The archiver certifies provenance and integrity of capture, not the truth of message content.